Software Requirements Specification (SRS)

for

SamaRasa (AI-Powered Companion System)

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1. Introduction

1.1 Purpose

The purpose of this document is to specify the comprehensive software requirements for **SamaRasa**, a bilingual AI-powered companion and health management system tailored for the Malaysian elderly population. This SRS details the functional behaviors, external interfaces, performance attributes, and design constraints for the **Low-Fidelity Prototype** deliverable. It serves as the primary contractual document between the development team and the stakeholders (lecturers/supervisors).

1.2 Document Conventions

- **Requirement Style:** "The system shall..." indicates a mandatory requirement. "The system should..." indicates a desirable requirement.
- **Prioritization Scheme:**
  - **[High]:** Critical core functionality; system is useless without it.
  - **[Medium]:** Important functionality; workaround exists but is painful.
  - **[Low]:** Nice-to-have features; can be deferred to Version 2.0.
- **Numbering:** Requirements are labeled REQ-<Module>-<ID> (e.g., REQ-MED-01).

1.3 Intended Audience and Reading Suggestions

- **System Architects:** Refer to **Section 2.1** and **Section 3** for system boundaries and API definitions.
- **UI/UX Designers:** Pay strict attention to **Section 2.3** and **Section 3.1**, which detail specific accessibility standards (WCAG 2.1) for the elderly.
- **Developers:** Focus on **Section 4** for detailed logic flows, exception handling, and data requirements.
- **Testers:** Use **Section 4** and **Section 5** to derive test cases and acceptance criteria.

1.4 Product Scope

SamaRasa aims to mitigate the challenges of Malaysia's aging society (7.3% aged 65+) by bridging the gap between independent living and caregiver supervision.

1.4.1 In-Scope

- **Voice-First Interface:** Bilingual (Bahasa Malaysia/English) interaction for all primary functions.
- **Health Management:** Medication reminders and simple mood tracking.
- **Safety:** Emergency SOS alerts and Fall Risk Assessment questionnaires.
- **Caregiver Portal:** A web dashboard for monitoring adherence and receiving alerts.

1.4.2 Out-of-Scope (for Prototype)

- **Real-time Telemedicine:** Live video calls with doctors are not included.
- **Hardware Integration:** Direct connection to smartwatches or IoT sensors is simulated, not physical.
- **Payment Gateways:** No in-app purchases or subscription handling.

1.5 Definitions, Acronyms, and Abbreviations

- **Low-Fidelity Prototype:** A conceptual implementation (wireframes or semi-functional code) used to validate user flows without a full backend.
- **Wizard of Oz:** A testing technique where AI responses are simulated by a human or pre-set scripts during the demo.
- **HCI:** Human-Computer Interaction.
- **STT / TTS:** Speech-to-Text / Text-to-Speech.
- **PDPA:** Personal Data Protection Act (Malaysia).

1.6 References

1. SSW3001 Group Project Guidelines, Semester 1 2025/2026.
2. Ministry of Health Malaysia, *National Policy for Older Persons (Revised 2024)*.
3. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.

2. Overall Description

2.1 Product Perspective

SamaRasa is a self-contained software solution that simulates a Client-Server architecture:

- **Client (Elderly):** A tablet-optimized web app acting as the "Digital Companion."
- **Client (Caregiver):** A responsive web dashboard for configuration and monitoring.
- **Server (Simulated):** A Python/Node.js backend handling logic, data persistence (SQLite), and AI response generation.

2.2 Product Functions

The system functions are categorized into four modules:

1. **Companion Module:** Daily greetings, weather updates, and empathetic conversation.
2. **Adherence Module:** Scheduling and enforcing medication intake.
3. **Safety Module:** Emergency broadcasting and risk evaluation.
4. **Analytics Module:** Visualizing health trends for family members.

2.3 User Classes and Characteristics

2.3.1 User Class: The Elderly (Primary)

- **Demographics:** Age 65-85, living alone or semi-independently in Malaysia.
- **Physical Constraints:**
  - **Vision:** Presbyopia (requires large fonts) and reduced color perception.
  - **Motor:** Mild hand tremors (requires large touch targets).
  - **Cognitive:** Short-term memory decline (requires frequent reminders).
- **Technical Constraints:** Low digital literacy; afraid of "breaking" the device.
- **Language:** Fluent in conversational Malay or English (Manglish).

2.3.2 User Class: The Caregiver (Secondary)

- **Demographics:** Children (30-50s) or hired nurses.
- **Goal:** Peace of mind and efficient management of the elderly's health routine.
- **Technical Skills:** High; comfortable with complex dashboards and data entry.

2.4 Operating Environment

- **Client Device:** * Tablet (minimum 10-inch screen recommended for elderly).
  - Smartphone (backup).
- **Software Platform:** Modern Web Browser (Chrome v90+, Safari v14+).
- **Network:** Continuous Internet connection (Wi-Fi/4G) required for Voice API and Cloud Sync.

2.5 Design and Implementation Constraints

- **Simulation:** Due to the 14-week timeline, complex AI (LLM) processing will be mocked using rule-based logic or simulated APIs.
- **Language Support:** The UI must support instant language switching without reloading the page.
- **Cost:** The solution must be built using open-source tools (no paid API subscriptions).

2.6 Assumptions and Dependencies

- **Device Availability:** The elderly user has access to a dedicated tablet with a functional microphone and speaker.
- **Caregiver Availability:** Caregivers have smartphones to receive email/SMS notifications.
- **Environment:** The device is used in a relatively quiet home environment (for accurate voice recognition).

3. External Interface Requirements

3.1 User Interfaces

The UI design focuses on **Accessibility** and **Simplicity**.

- **General Layout:**
  - **Grid System:** Simple 2x2 or 1x3 grid for main menu items.
  - **Navigation:** A persistent "Home" button is always visible. No hidden side menus.
- **Visual Standards:**
  - **Font:** San Francisco or Roboto (Sans-serif). **Min size:** 24px for body, 36px for headers.
  - **Contrast:** WCAG AAA Compliance (Contrast ratio > 7:1).
  - **Color Coding:** Green for "Confirm/Good", Red for "Emergency/Stop", Blue for "Information".
- **Voice Interaction (VUI):**
  - **Visual Cue:** When the system is listening, a pulsing waveform animation shall be displayed.
  - **Latency Masking:** Display "Thinking..." or "Sedang fikir..." while processing voice data.

3.2 Hardware Interfaces

- **Microphone:** Used for STT (Speech-to-Text). System must request permission on first launch.
- **Speaker:** Used for TTS (Text-to-Speech). System must default to 80% volume on startup.

3.3 Software Interfaces

- **Database System:** SQLite (local) or Firebase Firestore (cloud) for syncing data.
- **OS APIs:** Integration with the browser's Web Speech API for recognition and synthesis.

3.4 Communications Interfaces

- **Protocol:** HTTPS for all data transmission.
- **Data Format:** JSON (JavaScript Object Notation).
  - *Example Medication Object:*


```json
{
  "med_id": "101",
  "name": "Amlodipine",
  "dosage": "5mg",
  "time": "08:00",
  "status": "pending"
}
```

4. System Features (Functional Requirements)

4.1 AI Chat Companion & Emotional Support

Description: An intelligent chatbot that acts as a virtual grandchild, initiating conversation to reduce loneliness.

- **Pre-conditions:** Device is connected to the internet; Microphone permission granted.
- **Post-conditions:** User's mood score is updated in the database based on conversation sentiment.

Functional Requirements:

- **REQ-CHAT-01:** The system shall initiate a conversation upon app launch based on time of day (e.g., "Good Morning"). [High]
- **REQ-CHAT-02:** The system shall capture audio input and convert it to text using the Web Speech API. [High]
- **REQ-CHAT-03 (Sentiment Analysis):** The system shall parse the user's text for keywords (e.g., "Sad", "Sakit", "Lonely", "Happy") to determine the response tone. [Medium]
- **REQ-CHAT-04 (Fallback):** If the confidence score of voice recognition is < 60%, the system shall ask the user to repeat. [Medium]
- **REQ-CHAT-05:** The system shall support a "Quiet Mode" where interaction is text-only. [Low]

4.2 Smart Medication Reminder System

Description: A critical health module that interrupts the user to ensure medication compliance.

- **Pre-conditions:** Caregiver has configured a valid schedule.
- **Post-conditions:** Medication status is logged as TAKEN, SKIPPED, or MISSED.

Functional Requirements:

- **REQ-MED-01:** The system shall trigger a full-screen modal alert at the scheduled time. [High]
- **REQ-MED-02:** The alert shall include: Medication Name, Photo of the pill, Dosage instruction, and large "TAKEN" / "SKIP" buttons. [High]
- **REQ-MED-03 (Audio Loop):** The system shall verbally announce the reminder every 30 seconds until interaction occurs. [High]
- **REQ-MED-04 (Escalation):** If the status remains PENDING for >15 minutes, the system shall simulate sending an SMS alert to the caregiver. [High]
- **REQ-MED-05 (Logging):** The system shall record the timestamp of the action to the Health Diary. [High]

4.3 One-Touch Emergency Alert (SOS)

Description: A fail-safe mechanism for immediate assistance.

Functional Requirements:

- **REQ-SOS-01:** The SOS button shall be fixed at the bottom-right or top-right of the screen, distinct from other controls. [High]
- **REQ-SOS-02 (False Alarm Prevention):** Activating SOS requires a "Long Press" (3 seconds) or "Double Tap". [High]
- **REQ-SOS-03 (Countdown):** A 5-second countdown with a loud siren sound shall precede the actual alert transmission. [Medium]
- **REQ-SOS-04:** The alert payload sent to the caregiver shall include: User ID, Timestamp, and Simulated GPS Location. [High]

4.4 Health Diary & Caregiver Dashboard

Description: A web portal for family members to view summarized health data.

Functional Requirements:

- **REQ-DASH-01:** The dashboard shall display a calendar view showing medication adherence (Green tick / Red cross) for the past 30 days. [Medium]
- **REQ-DASH-02:** The system shall allow caregivers to remotely update the medication schedule (Add/Edit/Delete). [High]
- **REQ-DASH-03:** The dashboard shall visualize the user's "Mood Trend" (e.g., Line graph of reported happiness) over the week. [Low]

4.5 Fall Risk Assessment Module

Description: A digitized version of the CDC "STEADI" brochure.

Functional Requirements:

- **REQ-RISK-01:** The system shall present a questionnaire wizard with 5-10 Yes/No questions. [Low]
- **REQ-RISK-02:** The system shall calculate a risk score (0-10). Scores > 4 trigger a recommendation to "Consult a Doctor". [Low]

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **Latency:** Voice recognition to text display latency shall not exceed **2 seconds** under normal network conditions.
- **Throughput:** The system shall handle up to 100 concurrent simulated users for the demo.
- **Startup:** The application shall be fully interactive within **3 seconds** of launching.

5.2 Safety Requirements

- **Data Persistence:** In case of battery failure, medication logs must be saved to non-volatile storage (LocalStorage/DB) immediately upon entry.
- **Medical Disclaimer:** A disclaimer stating "This is a software aid, not a medical device" must be visible on the "About" screen.

5.3 Security & Privacy Requirements

- **Authentication:** Caregiver access requires a secure login (Email/Password).
- **Access Control:** The "Settings" menu on the Elderly device shall be PIN-protected to prevent accidental deletion of configurations.
- **Encryption:** All data at rest (simulated) and in transit (HTTPS) shall be encrypted.

5.4 Software Quality Attributes

- **Reliability:** The Medication Alarm feature must have a 99.9% success rate during testing.
- **Usability:** Targeted System Usability Scale (SUS) score > 75. Interface text must support 200% zoom without breaking layout.
- **Robustness:** The system must gracefully handle network disconnections by caching outgoing data and syncing when the connection is restored.

5.5 Legal and Regulatory Compliance

- **PDPA Compliance:** The system shall store minimal personal data. Users must explicitly consent to data collection during onboarding, in compliance with the Personal Data Protection Act 2010 (Malaysia).

Appendix A: Analysis Models

A.1 Use Case Diagram (Text Description)

- **Actor: Elderly User**
  - Uses -> **Chat with AI**
  - Uses -> **Confirm Medication**
  - Uses -> **Trigger SOS**
  - Uses -> **Log Daily Mood**
- **Actor: Caregiver**
  - Uses -> **Manage Medication Schedule**
  - Uses -> **View Health Reports**
  - Uses -> **Receive Alerts**

A.2 State Transition Diagram (Medication)

Pending -> (Time Reached) -> Alarming -> (User Action: Take) -> Taken
Pending -> (Time Reached) -> Alarming -> (User Action: Skip) -> Skipped
Pending -> (Time Reached) -> Alarming -> (Time > 15m) -> Missed

Appendix B: Data Dictionary (Simplified)

Entity	Attribute	Type	Description
User	userID	UUID	Unique identifier
	preferredLang	String	'en' or 'ms'
Medication	medID	UUID	Unique identifier
	name	String	Name of medicine
	scheduleTime	Time	HH:MM format
Log	logID	UUID	Unique identifier
	status	Enum	Taken/Skipped/Missed
	timestamp	DateTime	Exact time of event

Appendix C: To Be Determined (TBD) List

- **TBD-01:** Final decision on whether to use Google Cloud Speech-to-Text or built-in Web Speech API (pending cost analysis).
- **TBD-02:** Specific wording for the Malay translation of the Fall Risk Questionnaire.